

**This assignment will not be graded and is only for practice.**

**Note:** Suppose  $W_1$  and  $W_2$  are subspaces of a vector space  $V$ . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that  $W_1 + W_2$  is a subspace of  $V$ .

**Multiple Choice Questions (MCQ):**

1) An inner product on  $\mathbb{R}^3$  is defined as:

**1 point**

$$\langle \cdot, \cdot \rangle: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

Match a set  $S$  in Column A with the set  $T$  given in column B, so that  $T \subset \text{Span}(S)^\perp$ . Match a set in column B with the property it satisfies in column C.

|    | Set ( $S$ )                |      | set ( $T$ )                                                     |    | Properties                                                                                    |
|----|----------------------------|------|-----------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------|
|    | (Column A)                 |      | (Column B)                                                      |    | (Column C)                                                                                    |
| a) | $\{(-2, 1, 3)\}$           | i)   | $\{(0, 1, 1), (2, -1, 1)\}$                                     | 1) | forms an orthonormal basis of a vector space which is isomorphic to $\mathbb{R}^2$            |
| b) | $\{(1, -2, 1)\}$           | ii)  | $\{(1, -1, 1), (2, 1, 1)\}$                                     | 2) | orthogonal but not orthonormal                                                                |
| c) | $\{(1, 1, -1)\}$           | iii) | $\{\frac{1}{\sqrt{2}}(0, -1, 1)\}$                              | 3) | not orthogonal                                                                                |
| d) | $\{(0, 1, 1), (1, 1, 1)\}$ | iv)  | $\{\frac{1}{\sqrt{3}}(1, 1, 1), \frac{1}{\sqrt{2}}(-1, 0, 1)\}$ | 4) | orthonormal but does not form a basis of a vector space which is isomorphic to $\mathbb{R}^2$ |

Table : M2W9P1

Which of the following option are true?

☐ d  $\rightarrow$  ii  $\rightarrow$  3

☐ d  $\rightarrow$  iii  $\rightarrow$  4

☐ c  $\rightarrow$  ii  $\rightarrow$  4

☐

a  $\rightarrow$  iv  $\rightarrow$  1

☐ a  $\rightarrow$  ii  $\rightarrow$  3

☐ b  $\rightarrow$  iv  $\rightarrow$  2

No, the answer is incorrect.

Score: 0

Accepted Answers:

d  $\rightarrow$  iii  $\rightarrow$  4

a  $\rightarrow$  ii  $\rightarrow$  3

2) Suppose  $W_1$  and  $W_2$  are subspaces of a vector space  $V$ . Let  $P_{W_1}$  and  $P_{W_2}$  denote the projection from  $V$  to  $W_1$  and  $V$  to  $W_2$  respectively and consider the following statements:

**Statement P:** If  $P_{W_1} + P_{W_2}$  is a projection from  $V$  to  $W_1 + W_2$ , then  $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$ .

**Statement Q:**  $P_{W_1}^2 := P_{W_1} \circ P_{W_1}$  is not a projection from  $V$  to  $W_1$ .

**Statement R:** If  $A$  is the matrix representation of  $P_{W_2}$ , then  $A$  can not be a symmetric matrix.

**Statement S:**  $P_{W_1} - P_{W_2}$  is a projection from  $V$  to  $W_1 + W_2$ .

Find the number of correct statements

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

3) Which of the followings option(s) is (are) true?

**1 point**

☐ Let  $u, v$ , and  $w$  be three vectors in  $\mathbb{R}^2$  and  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}^2$ . If  $\langle u, v \rangle = \langle u, w \rangle$ , then  $v = w$

☐ Let  $V$  be an inner product space. If  $\langle u, v \rangle = 0$  for all  $u \in V$ , then  $v = 0$

- ☐ Let  $V$  be an inner product space and  $u, v \in V$ . If  $\langle u + v, u - v \rangle = 0$ , then  $\|u\| = \|v\|$
- ☐ Let  $V$  be an inner product space. If  $S \subset V$  then  $S^\perp = (\text{span}(S))^\perp$
- ☐ Suppose  $\{u_1, u_2, \dots, u_m\}$  be an orthogonal set of vectors in an inner product space  $V$ , then  $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$ .

No, the answer is incorrect.

Score: 0

#### Accepted Answers:

Let  $V$  be an inner product space. If  $\langle u, v \rangle = 0$  for all  $u \in V$ , then  $v = 0$

Let  $V$  be an inner product space. If  $S \subset V$  then  $S^\perp = (\text{span}(S))^\perp$

Suppose  $\{u_1, u_2, \dots, u_m\}$  be an orthogonal set of vectors in an inner product space  $V$ , then  $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$ .

4) Which of the following option(s) is/are true?

**1 point**

- ☐ Let  $A$  and  $B$  be orthogonal matrices of order 3. Then  $A$  and  $B$  are equivalent matrices.
- ☐ Let  $A$  and  $B$  be orthogonal matrices. Then  $A$  and  $B$  are similar matrices.
- ☐ Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle  $\theta$ . Then nullity of the matrix  $A$  is 0.
- ☐ Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle  $\theta$  and  $B$  be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X - axis with angle  $\beta$ . Then  $A$  and  $B$  are equivalent matrices.

No, the answer is incorrect.

Score: 0

#### Accepted Answers:

Let  $A$  and  $B$  be orthogonal matrices of order 3. Then  $A$  and  $B$  are equivalent matrices.

Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle  $\theta$ . Then nullity of the matrix  $A$  is 0.

Let  $A$  be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle  $\theta$  and  $B$  be a rotation matrix corresponding to the anti- clock wise

rotation of the  $YZ$ -plane about the  $X$ -axis with angle  $\beta$ . Then  $A$  and  $B$  are equivalent matrices.

**Numerical Answer Type:**

5) Let  $u = (1, 2, 1)$  and  $v = (1, 2, 3)$  be the vectors of the inner product space  $\mathbb{R}^3$  with usual inner product. Let  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be an orthogonal linear transformation. If  $\theta$  is an angle between  $T(u)$  and  $T(v)$ , then find the value of  $||T(u)|| ||T(v)|| \cos \theta$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

**1 point**

6) Consider a set  $W = \{(1, 1, 1)\}$  from the inner product space  $\mathbb{R}^3$ . Find the dimension of the subspace  $W^\perp$ , where  $W^\perp$  is the collection of the vectors which are orthogonal to the vector  $(1, 1, 1)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

7) Let  $v = (3, 1, 2)$  be a vector in  $\mathbb{R}^3$ . If  $(a, b, c)$  is the vector obtained from  $v$  after the anti-clock wise rotation of ZX-plane with angle  $45^\circ$  about the Y- axis, then find the value of  $\sqrt{2}(a + b + c - 1)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

**1 point**

**Comprehension Type Question:**

With a particular frame of reference (in  $\mathbb{R}^2$ ) the position of Rahul's house is at the point  $(8,1)$ . There are two roads Road 1 and Road 2 along the lines  $y = 2x$  and  $y = -2x$  respectively on the  $XY$ -plane.

Answer questions 8,9 and 10 using the given information.



8) Which of the following option(s) is (are) true?

**1 point**

- ☐ If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
- ☐ If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at  $\frac{6}{5}(1, -2)$ .
- ☐ If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at  $\frac{6}{5}(1, -2)$ .
- ☐ If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

No, the answer is incorrect.

Score: 0

Accepted Answers:

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at  $\frac{6}{5}(1, -2)$ .

9) Let  $\theta$  be the acute angle between Road 1 and Road 2, then what will be the value of  $5 \cos(\theta)$ ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

**1 point**

10) Let  $\langle \cdot, \cdot \rangle$  denote the standard inner product on  $\mathbb{R}^2$ , i.e.,  $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 + x_2y_2$ . Suppose  $W_1 = \{(x, y) \mid y = 2x\}$  and  $W_2 = \{(x, y) \mid y = -2x\}$  are two subspaces of  $\mathbb{R}^2$ . Which of the following options is correct?

**1 point**

- ☐  $W_1^\perp = \{(x, y) \mid 2y = x\}$

☐  $W_2^\perp = \{(x, y) \mid 2y = -x\}$

☐  $W_1^\perp = \{(x, y) \mid 2y = -x\}$

☐  $W_2^\perp = \{(x, y) \mid 2y = x\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$W_1^\perp = \{(x, y) \mid 2y = -x\}$

$W_2^\perp = \{(x, y) \mid 2y = x\}$

Your score is: 0/10